

Solved Paper	Maharashtra HSC Exam March 2017	Biology
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Time : 3 Hours

Max. Marks: 70

General Instructions:

- All questions are compulsory.
- Draw neat and labelled full marks.
- Figures to the right indicate full marks.
- Answer to the question in Section –I and Section –II must be written two separate Answer books.
- Questions form Section –I attempted in the Answer book of Section –II and vice – versa will be assessed/not be given any credit.
- Answer to every new question must be started on a new page.

SECTION – I

1. Select and write the most appropriate Answer form the given alternatives (along with its alphabets) for each sub-question: [7]

(i) The genotype of human blood group B is _____

- (a) $I^A i$ (b) $I^B i$ (c) $I^A I^A$ (d) ii

Answer. $I^B i$

(ii) Breakdown of detritus into smaller particles is called

- (a) Fragmentation (b) leaching
(c) catabolism (d) humiliation

Answer. (b) Leaching

(iii) In Brassica (rapeseed , mustard) _____ variety is resistant to Aphids.

- (a) Pusa A-4 (b) Pusa Gaurav
(c) Pusa Sawni (d) Pusa Shubra

Answer. (b) Pusa Gaurav

(iv) The antibiotic chloromycetin is obtained from _____.

- (a) Sclerotinia libertina (b) Aspergillus niger
(c) Streptomyces griseus (d) Streptomyces venezuelae

Answer. (d) Streptomyces venezuelae

(v) The _____ enzyme is used to cut DNA at specific point.

- (a) DNA polymerase (b) Alkaline phosphatase
(c) Restriction endonuclease (d) DNA ligase

Answer. (c) Restriction endonuclease

(vi) R.Q. for proteins is about

- (a) 0.7 (b) 0.8
(c) 0.9 (d) 1.0

Answer. (b) 0.8

(vii) Ozone depletion is occurring widely in the stratosphere, it leads to ozone hole caused mainly due to _____.

- (a) Ethylene (b) methane
(c) CFCs (d) CO₂

Answer. (c) CFCs

2. (A) Answer each question in 'one' sentence only. (6)

(i) Give an example of the source of thermostable enzyme DNA polymerase.

Answer. Pfu DNA polymerase.

(ii) Give an example of the non-edible or poisonous mushroom, studied by you.

Answer. Amanita phalloides.

(iii) Name the secondary metabolites in Catharanthus roseus.

Answer. Indole alkaloids

(iv) What is meant by ecological succession?

Answer. Ecological succession is the process of change in the species structure of an ecological community over time. The time scale can be decades (for example, after a wildfire), or even millions of years after a mass extinction.

(v) Name the organism and enzyme which bring about alcoholic fermentation of sucrose.

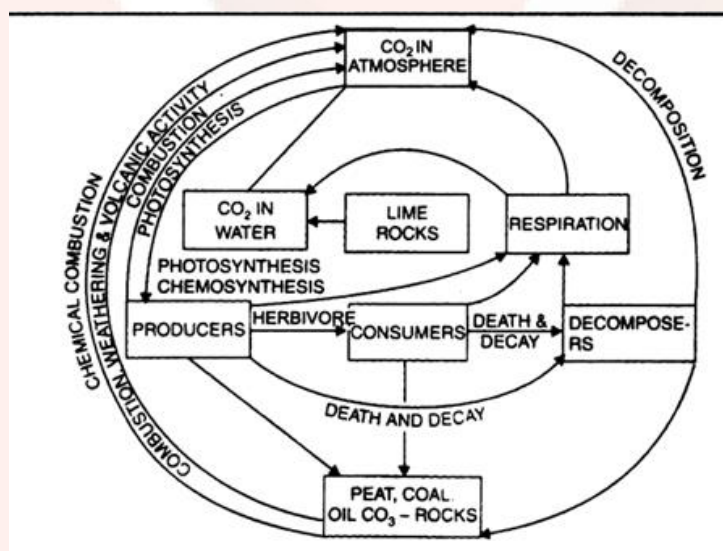
Answer. Flowers have large brightly coloured petals.

(vi) Enlist any 'two' floral adaptations in salvia.

Answer. The calyx is normally tubular or bell shaped, without bearded throats, and divided into two parts or lips, the upper lip entire or three-toothed, the lower two-cleft.

(B) Given schematic representation of carbon cycle. (2)

Answer.



(c) Attempt any TWO of the following: (4)

(i) What is a 'test cross'? Explain significance of a test cross.

Answer Test cross is a method devised by Mendel to verify (test) the genotype of F_1 (or F_2/F_3) hybrids. In test cross F_1 hybrid (or F_2) is crossed with homozygous recessive parent. Since the offspring is crossed back with one of the parents, it is also called cross.

F_2 generation of test cross consists of 50% heterozygous tall plants and 50% homozygous dwarf plant.

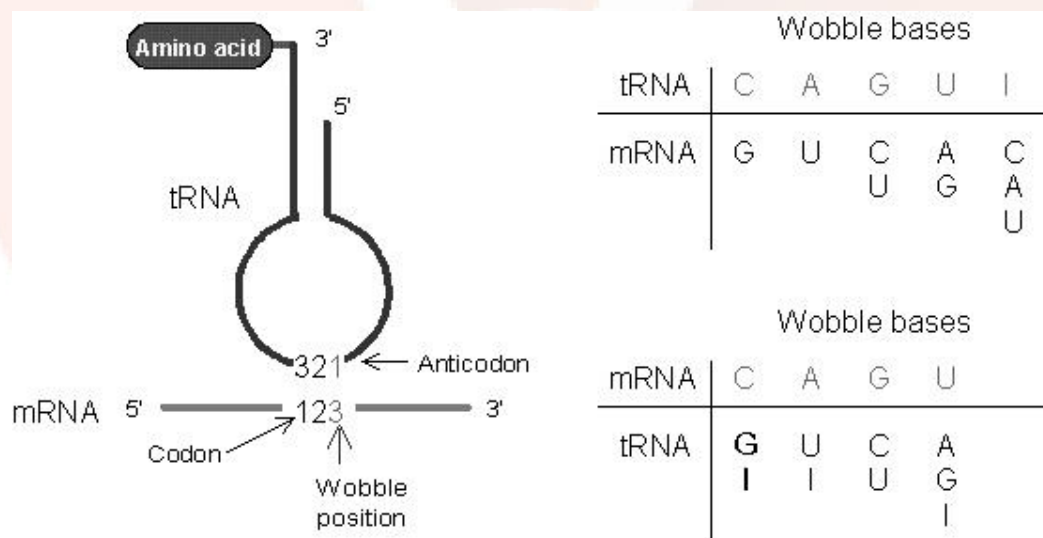
Significance of test cross:

- (a) Test cross can be used to find out genotype of any plant with dominant expression.
- (b) It is also used to introduce useful recessive trait in the hybrids in rapid crop improvement programmes.
- (c) It is more popular with breeders and geneticists.
- (d) It is a simple method to verify (test) the genotype of F_1 hybrids.
- (e) Test cross is of great help in the production of line varieties.

(ii) Explain 'Wobble hypothesis' with the help of a suitable diagram.

Answer. Wobble hypothesis was proposed by Crick in 1966. According to this hypothesis, the nitrogen base present at the first position of the anticodon on tRNA is generally abnormal. These bases have the capability to pair up with more than one type of nitrogen base of the codon of mRNA (at third position). This is also called Wobble base pair hypothesis.

The Wobble hypothesis proposes that normal base pairing can occur between nitrogen bases in positions 1 and 2 of the codon and the corresponding bases (3 and 2) in the anticodon. Actually, the base 1 in anticodon can form non-Watson-Crick base pairing with the third position of the codon.



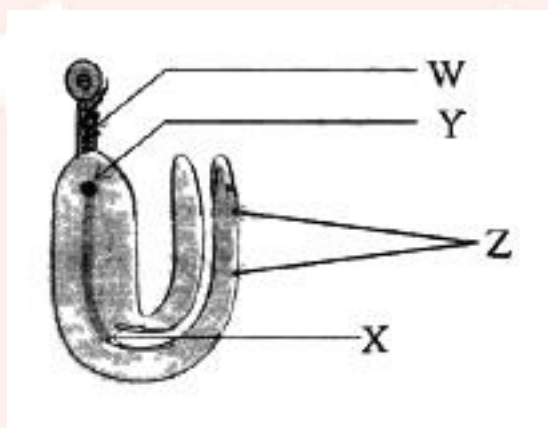
(iii) What is a 'biopatent'? Explain it with a suitable example.

Answer: Patents granted in the field of biology are called as bio-patents. Patents on plants, microorganisms can be categorised as bio- patents.

Bio patent refers to patents obtained by individuals/ groups/country to establish exclusive right over the origin of a particular organisms/ plant. Such right gives the holder to claim global trading rights- without any other group claiming the trading right.

For example: An American group had claimed global patent over basmati rice originated in India. It was successfully fought at WTO level and India obtained patent on it.

(iv) Name the parts W,X,Y and Z from the following figures:



Answer:

X--- Plumule

Y-----Radicle

Z-----Cotyledons

W-----Suspensor

3. (A) Attempt any TWO of the following

(6)

(i) Explain replication of bacteriophage with the help of a suitable diagram.

Answer: A bacteriophage is a virus that infects bacteria. They are commonly known as 'phage'.

Lytic cycle i.e replication of bacteriophages takes place inside the specific host / bacterial cell. It takes place in following steps:

- (a) Attachment: Bacteriophage attach to specific receptors on the surface of bacteria. As phages do not move independently, they rely on random encounters with the right receptors.
- (b) Penetration: After the contact the tail fibres brings the base plate closer to the surface of the cell. On complete attachment the tail contracts, injecting DNA through the bacterial membrane.
- (c) Synthesis of proteins and nucleic acid: The host's normal synthesis of proteins and nucleic acid is disrupted and it is forced manufacture viral DNA, and proteins instead.
- (d) Virion assembly: The base plates are assembled with the tail first. The heads-capsid are constructed separately and then are joined with the tails. DNA is packed efficiently within the heads. The whole process takes about 15 minutes.
- (e) Release of virions: Phages are released via lysis of cell. Endolysin breaks down the cell wall. Released virions are capable infecting a new bacterium.

(ii) What are bio fertilizers? Explain them with suitable example

The fertilizer that uses bio-agents like culture of blue-green algae or other micro-organisms instead of chemicals in order to make soil fertile and replenish its nutrients is known as bio-fertilizer. The bio agents that are used are Rhizobium, Anabena, Nostoc, Azotobacter, Pseudomonas, Blue green algae.

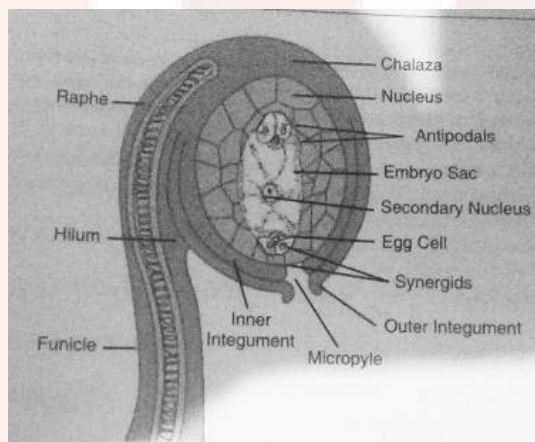
Rhizobium, Azotobacter, Azospirillum and Blue green algae (BGA) have been traditionally used Biofertilizers. Rhizobium inoculants are used for leguminous crops such as pulses. Azotobacter can be used with crops like wheat, maize, mustard, cotton, potato and other vegetable crops.

(iii) Differentiate between anemophily and entomophily.

S.No.	Anemophily	Entomophily
1	Flowers small and inconspicuous	Flowers big and showy
2	No nector production	Nector production
3	Petals reduced or absent	Petals big and colourful
4	Inflorescences with many flowers	Inflorescences with many flowers
5	Long and exposed stamens	Short stamens
6	Long and feather – like stigmas	Short and flat stigmas
7	Usually unisexual flowers	Usually bisexual flowers
8	Pollen produced in large quantities	Pollen produced in smaller quantities
9	Light and dry pollen	Heavier and sticky pollen
10	Pollination isn't specialized	Pollination is specialized
11	Typical of Fagales (trees) and Poales(grasses)	Majority of flowering plants

(B) Sketch and label V.S. of mature anatropous ovule. [3]

Answer:



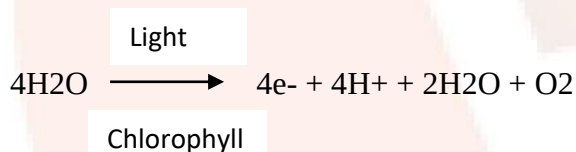
4. What is 'photophosphorylation'? Describe non-cyclic photophosphorylation with schematic representation. Give its significance.

Answer: The process in which energy rich ATP molecules are synthesized using solar energy is called photophosphorylation. Non-cyclic photophosphorylation involves photolysis of water and a continuous unidirectional flow of electrons from water to PSII, from to PSI via cytochromes and finally to NADP. It takes place in the following steps:

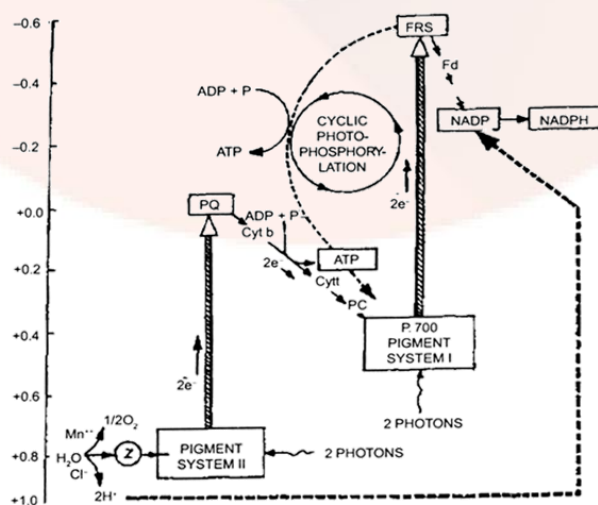
Photoexcitation of PSII and flow of electrons from PSII to PSI: The pigments in PSII absorb energy of different wavelengths of light which is funneled to the reaction centre P680 or chl-a 680. Energy rich electrons are expelled from p680. These electrons are taken up by the plastoquinons (PQ) through coenzyme quinine (CO-Q) leaving the reaction centre ionized and PQ gets reduced. The energy rich electrons from PQ roll down the chain of various electron carriers. Components of electron transport chain are cytochromes b6, cytochromes f and plastocynin (PC). The reduced PC transfers the de-energized electrons to the reaction centre of PSI i.e P700.

Photoexcitation of PSI: The pigments in PSI absorb the energy of different wavelengths of light which is funnelled to the reaction centre i.e, chl-a 700 or P700. The energy rich electrons are expelled from excited chl-a molecule and these are taken up by iron containing red protein ferredoxin (Fd) through an unknown electron acceptor which is called ferredoxin reducing substance (FRS). From reduced ferredoxin these electrons are taken up by coenzyme NADP, NADP retains the electrons and becomes negatively charged NADPH. PSII replaces the electrons in PSI.

Photolysis of water: P680 + acts as a strong oxidizing agent and brings about splitting of water molecules to release oxygen. This light dependent splitting of water molecule is called photolysis of water.



The four electrons generated in breakdown of four water molecules are accepted by oxidised P680. The protons (4H^+) are accepted by 2NADP to form 2NADPH_2 in presence of ferredoxin NADP reductase.



OR

Q 4. What is 'RNA'? Explain different types of non-genetic RNA with diagrams and functions.

Answer: RNA is a ribonucleic acid found in the nucleus as well as in the cytoplasm. It is a single stranded structure which may be simple and straight or may be variously folded or coiled upon itself. It is made up to ribose sugar, phosphate as phosphoric acid and n-bases. They are of mainly two types:

Genetic RNA which acts as genetic material of some viruses.

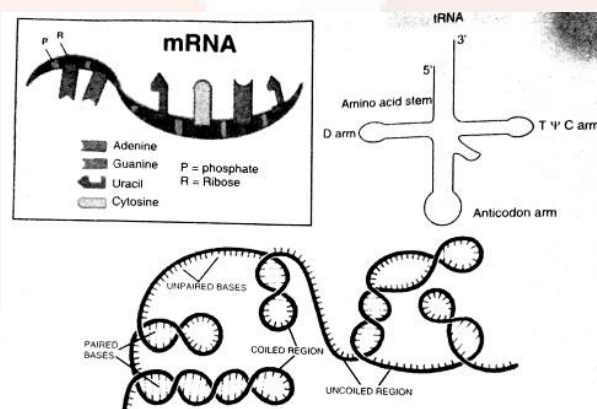
Non-genetic RNA function for the protein synthesis in most of the organisms. They are of three types:

m-RNA or Messenger RNA : It carries coded message for protein synthesis from DNA to ribosomes about the sequence of amino acids in the polypeptide chains.

r-RNA or Ribosomal RNA : It provides proper binding site for m-RNA . It orients the m-RNA molecule in such a way that all the codons are properly read.

t-RNA or Transfer RNA : It carries specific type of amino acid at CCA end to ribosomes during protein synthesis.

It places the required amino acid properly in the sequence and translates the coded message of m-RNA in terms of amino acids.



SECTION II

ZOOLOGY

Q 5. Select and write the most appropriate Answer from the given alternatives (along with its alphabets) for each sub question: [7]

(1) Which of the following has normal vision?

- (a) $X^c X^c$ (b) $X^c y$
- (c) $XC Xc$ (d) $X^c yc$

Answer. (b) $X^c y$

(2) In DNA fingerprinting technique, radioactive DNA probe is obtained from _____ of female banded Krait snake.

- (a) X chromosome (b) Y chromosome
- (c) X and Y chromosome (d) Autosome

Answer. (b) Y chromosome

(3) Abortion in the first trimester of pregnancy may occur due to lack of _____.

- (a) Aldosterone (b) testosterone
- (c) Oestrogen (d) progesterone

Answer. (d) progesterone

(4) _____ contributes about 60% of the total volume of the semen.

- (a) Prostate gland (b) Cowper's gland
- (c) Seminal vesicles (d) Bartholin's glands

Answer. (c) Seminal vesicles

(5) Lowering of blood pressure is related with the production of_____.

- (a) ADH (b) ANF
(c) GH (d) LH

Answer. (b) ANF

(6) Humulin is used to treat _____.

- (a) Diabetes mellitus (b) Diabetes insipidus
(c) Hepatitis (d) Nephritis

Answer. (b) Diabetes insipidus

(7) The modification of original genetic make up is focussed by_____.

- (a) PCR (b) DNA fingerprinting
(c) Electrophoresis (d) Gene therapy

Answer. (d) Gene therapy

Q 6. (A) Answer the following questions only in 'one' sentence each: (6)

(i) Which material is used is used for isolation of DNA in fingerprinting technique?

Answer. DNA is isolation forms any material available like blood, semen, hair roots, tissue sample etc.

(ii) Give significance of podocyte.

Answer. Podocytes play an important role in glomerular function. Together with endothelial cells of the glomerular capillary loop and the glomerular basement membrane they form a filtration barrier.

(iii) What is 'commensalism'?

Answer. Commensalism is an association between two organisms in which one benefits and other derives neither benefit nor harm.

(iv) What is the function of acrosome?

Answer. In Eutherian mammals the acrosome contains digestive enzymes (including hyaluronidase and acrosin) which break down the outer membrane of the ovum, called the zone pellucida, allowing the haploid nucleus in the sperm cell to join with the haploid nucleus in the ovum.

(v) Distinguish between X and Y chromosome. (Mention any 'two' points.)

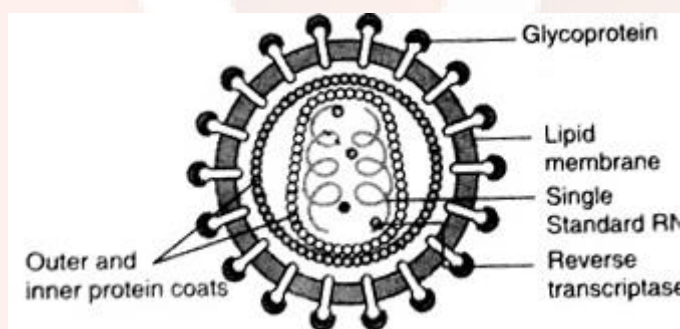
Answer. Males have one Y chromosome and one X chromosome, while females have two X chromosome. In mammals, the Y chromosome contains a gene SRY, which triggers embryonic development as a male. The X chromosome contains over 1000 genes, compared to 78 genes on the Y chromosome.

(vi) Give any 'two' examples of endangered species.

Answer. Javan rhino (*Rhinoceros sondaicus*) and Blue Throated Macaw.

(B) Sketch and label the 'Structure of HIV'. (2)

Answer. Structure of HIV



(c) Attempt any Two of the following: (4)

(i) Write a note on erythrocytes.

Answer. There are important cells in your body that travel in blood. They are involved in a gas exchange that is essential to human life. Red blood cells (RBCs) are their most common name, but they are also called erythrocytes.

Erythrocytes are round. When these cells are normal they can look like doughnuts with the holes in the center. Haemoglobin is responsible for erythrocytes round shape; it increases their surface area, allowing them to carry more oxygen molecules. Eerythrocytes are created in the bone marrow through a process called erthropoiesis, before being released into the blood stream . Their life span is 120 days.

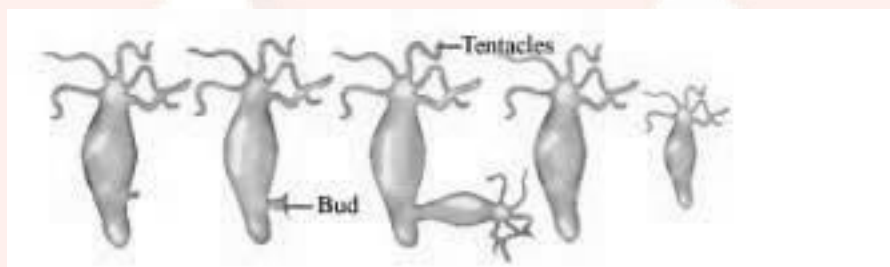
(ii) What are the uses of vaccine?

Answer. Applications of vaccine:

- (a) Vaccine is used to stimulate the production of antibodies.
- (b) It helps to induce immunity against several diseases.
- (c) It helps to protect against disease.

(iii) Describe the process of budding in hydra.

Answer. In Hydra, the cells divide at a specific site and develop as an outgrowth called a bud. These buds, while attached to the parent plant, develop into small individuals. When this individual becomes large enough, it detaches itself from the parent body to exist as an independent individual.



(iv) Name the species used in sericulture. Name the stages in the life cycle of a moth in cyclic form.

Answer. Bombyx mori is used sericulture.

Stages in life cycle are: Eggs (on mulberry leaf) → caterpillar → cocoon → silk moth.

Q 7. (A) Attempt any TWO of following: (6)

(i) Explain ABO blood group system in human being with a suitable chart.

Answer: Inheritance pattern of ABO blood groups in human show dominance and multiple allelism can be explained as:

- a. In dominance, F₁ progeny resembles one of the parent
- b. In codominance, the F₁ progeny resembles both the parents
- c. Multiple alleles: When more than two alleles control a character, as in human blood groups.
- d. ABO blood groups are controlled by gene 1. Gene 1 has these alleles, 1^A, 1^B and i. A person possesses any two of three alleles.
- e. I and I dominate over i. But with each other. 1^A and 1^B are codominant.
- f. 1^A and 1^B contain A and B types of sugar, while i does not contain any sugar.

Blood type	Antigen on red blood cells	Antibodies in plasma	Can receive Blood from	Can donate Blood to
A	A	anti-B	O and A	A and AB
B	B	anti-A	O and B	B and AB
AB	A and B	neither	O,A,B And AB	AB only
O	neither	anti-B and anti-A	O only	O,A,B And AB

(ii) Describe diagrammatic representation of age structure showing declining populations.

Answer:

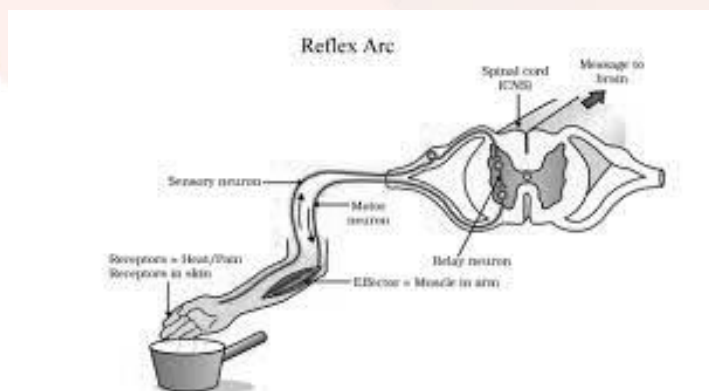
It has a large number of post reproductive and smaller number of pre reproductive population.

(iii) With the help a neat and labelled diagram, describe reflex arc.

Answer: When we accidentally touch a hot object, we withdraw our hands immediately without thinking. If we do not do this, our hands will burn.

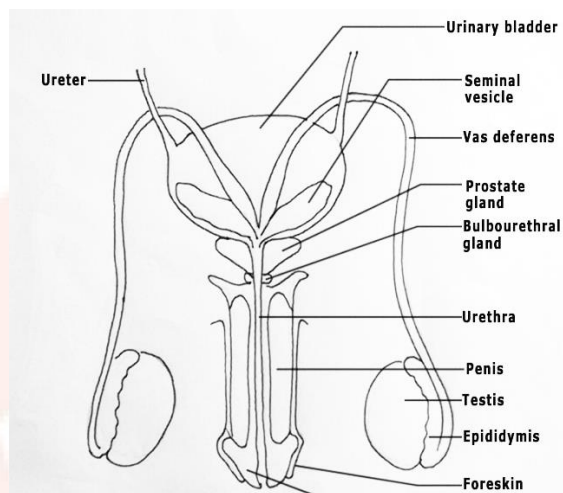
The sensory nerves detect the heat. They are connected to the nerves, which move the muscles of the hand. Such a connection of detecting the signal from the nerves(input), and responding to it immediately (output) is called a reflex arc.

A reflex arc makes instant and automatic responses possible. It connects the input nerve and output nerve, and meets in a bundle in the spinal cord. In fact, nerves from all over the body meet in a bundle in the spinal cord, on their way to the brain. Therefore, the information input reaches the brain.



(B) Sketch and label 'human male reproductive system' (3)

Answer:



8. Enlist human endocrine glands. Describe the TS of thyroid gland and a note on deficiency of thyroxine. [7]

Answer:

- (a) Adrenal
- (b) Hypothalamus
- (c) Pineal
- (d) Pituitary
- (e) Thyroid
- (f) Thymus
- (g) Pancreas
- (h) Testes
- (i) Ovaries

The thyroid gland is the largest endocrine gland located anterior to the thyroid cartilage of the larynx in the neck. The gland is well supplied with blood vessels. It is a bilobed organ. The two lobes are connected by a narrow structure called the isthmus. The microscopic structure of the thyroid gland shows thyroid follicles composed of cubical epithelium and filled with a homogenous material called colloid. Small amount of loose connective tissue forms stroma of the gland. Besides containing blood capillaries, the stroma contains small clusters of specialised Para follicular cells or 'C' cells. The thyroid gland is the only gland that stores hormones in large quantity for about two months.

Deficiency of Thyroxin

- (a) Excessive secretion of thyroid hormone (hyperthyroidism) results in exophthalmic goiter or Grave's disease. It is accomplished by the bulging of the eye ball. It is associated with high metabolic rate, heart beat and blood pressure rises, restlessness, tremors, nervousness, rise in body temperatures etc.
- (b) Less secretion of thyroid hormone (hypothyroidism) results in Myxedema in adults and cretinism (feeble mindedness) in children. The patient of myxedema suffers from low metabolic rate, slow heart rate, low body temperature and reproductive failure.
- (c) For the synthesis of thyroid hormone (thyroxin), an important inorganic ion called iodine is needed in the body. Hence the dietary deficiency of iodine causes goiter in which thyroid gland enlarges in an effort to produce more thyroxine.
- (d) Failure of thyroid hormone secretion in children slows body growth and mental development and reduces metabolic rate. The child becomes stunted and mentally retarded. The body temperature, heart rate and blood pressure are lower than normal. The patient is pot-bellied and pigeon chested and has a protruding tongue. This disease is known as cretinism.

OR

Define 'evolution'. Give the principles of Darwin's theory of natural selection. Mention any 'one' objection to it.

Answer:

Evolution is change in the heritable characteristic of biological populations over successive generations. Evolutionary processes give rise to biodiversity at every level of biological organisation, including the levels of species, individual organisms, and molecules.

Darwin's theory of organic evolution by natural selection is based on following principles:

- (a) Over production of enormous fertility or prodiagality of production: Every living organism has a great potential for reproduction. All organisms produces more young ones than those which can survive up to maturity. Any population tends to increase in geometrical proportion.
- (b) Struggle for existence: Individuals multiply in geometric ratio but the spoace and food remain constant. This leads to competition for getting requirements of life. This competition is called as struggle foe existence.
- (c) Variation and heredity: variations are the differences between individuals of the same species. Variations may be harmful, neutral or useful. The useful variations are presented and passed on to the offsprings. They play an important role in evolution.

- (d) Survival of the fittest or natural selection: The organisms provided with favourable variations succeed in the struggle for existence. While the organisms with unfavourable variations are unfit to survive and they perish. This means survival of the fittest.
- (e) Origin of new species: Due to struggle for existence, variability and inheritance, the organisms tend to become better adapted to their environment. The favourable characters are transmitted to the next generations. In the succeeding generation also natural selection operates. Hence in each and every generation, there is slight betterment of the already developed characters.

Objectives to Darwin's theory of Natural Selection: The objectives raised against Darwin's theory of Natural Selection are as follows:

- (a) Darwin could not explain 'the arrival of the fittest'. He simply insisted on survival of the fittest.
- (b) He did not consider the hereditary principles.
- (c) He did not differentiate between somatic and germinal variations
- (d) This theory has no satisfactory explanation for the cause, origin and inheritance of variations.
- (e) Darwin's theory of natural selection does not give any idea about the inheritance of useless variations like the vestigial organs.